

APPENDIX

VI. ROBOT DETAILS

A. Joints

Our Unitree G1 Edu+ consists of 27 joints with their names as mentioned in Table IV.

TABLE IV: Joints of the Unitree G1 Edu+ grouped by body part.

Legs	
left_hip_pitch_joint	right_hip_pitch_joint
left_hip_roll_joint	right_hip_roll_joint
left_hip_yaw_joint	right_hip_yaw_joint
left_knee_joint	right_knee_joint
left_ankle_pitch_joint	right_ankle_pitch_joint
left_ankle_roll_joint	right_ankle_roll_joint
Waist	
waist_yaw_joint	
(Left Right) Arms	
left_shoulder_pitch_joint	right_shoulder_pitch_joint
left_shoulder_roll_joint	right_shoulder_roll_joint
left_shoulder_yaw_joint	right_shoulder_yaw_joint
left_elbow_joint	right_elbow_joint
left_wrist_roll_joint	right_wrist_roll_joint
left_wrist_pitch_joint	right_wrist_pitch_joint
left_wrist_yaw_joint	right_wrist_yaw_joint
(Left Right) Hands	
left_hand_index_0_joint	right_hand_index_0_joint
left_hand_index_1_joint	right_hand_index_1_joint
left_hand_middle_0_joint	right_hand_middle_0_joint
left_hand_middle_1_joint	right_hand_middle_1_joint
left_hand_thumb_0_joint	right_hand_thumb_0_joint
left_hand_thumb_1_joint	right_hand_thumb_1_joint
left_hand_thumb_2_joint	right_hand_thumb_2_joint

B. Keypoints

The 41 keypoints on the robot are named as detailed in Table V.

C. PD control

We use PD controller to convert target joint angles to torque, τ_t as follows:-

$$\tau_t = k_p(q_t^{\text{commands}} - q_t^{\text{robot}}) - k_d\dot{q}_t^{\text{robot}}$$

Where k_p and k_d are positional and derivative gains and q_t^{commands} are the joint commands. The k_p, k_d gains for each joint are given in Table VI

VII. OPEN AND CLOSED STATES FOR EACH TASK

Each task has a specific set of joint angles for open and closed hand states to facilitate task-specific function. For instance, we want all fingers to open and closed in open and closed states of hand respectively while for button pressing, we only need the index finger open. For boxing, we keep all fingers closed for both open and closed as we never need to open them for the task. The exact finger joint angles for open and closed hand states for each task are listed in Table VII.

TABLE V: Keypoints of the Unitree G1 Edu+ grouped by body part.

Legs	
left_hip_pitch_link	right_hip_pitch_link
left_hip_roll_link	right_hip_roll_link
left_hip_yaw_link	right_hip_yaw_link
left_knee_link	right_knee_link
left_ankle_pitch_link	right_ankle_pitch_link
left_ankle_roll_link	right_ankle_roll_link
Waist & Torso	
pelvis	pelvis_contour_link
waist_yaw_link	waist_roll_link
torso_link	waist_support_link
logo_link	
Head & Sensors	
head_link	imu_link
d435_link	mid360_link
Arms	
left_shoulder_pitch_link	right_shoulder_pitch_link
left_shoulder_roll_link	right_shoulder_roll_link
left_shoulder_yaw_link	right_shoulder_yaw_link
left_elbow_link	right_elbow_link
left_wrist_roll_link	right_wrist_roll_link
left_wrist_pitch_link	right_wrist_pitch_link
left_wrist_yaw_link	right_wrist_yaw_link
left_rubber_hand	right_rubber_hand

VIII. KEYWORD-BASED FILTERING FOR FID

We use the following keywords to filter out trajectories that are used for FID calculation to evaluate human-ness of keypoint trajectories followed by **Ours** against *TaskOnly*:-

- **Pick**: All motions whose prompts contain “pick” keyword in them
- **Press Button**: All motions whose prompts contain “press” keyword and “button” keyword in them
- **Precise Punch**: All motions whose prompts contain “punch” keyword in them
- **Precise Kick**: All motions whose prompts contain “kick” keyword in them
- **Jump**: All motions whose prompts contain “jump” keyword in them

IX. REFERENCE TRAJECTORY GENERATION

For each task, we have a text prompt, λ^{text} and a spatial control signal $\lambda^{\text{spatial}} \in \mathcal{R}^{L \times S \times 3}$ where $L = 196$ is the no of time-step, $S = 22$ is the number of SMPL joints named as follows:-

TABLE VI: Joint list (unrolled column-wise) with default angle, K_p , and K_d all initialized to 0.

Joint name	Default angle	K_p	K_d
left_hip_pitch_joint	-0.2	200	5
left_hip_roll_joint	0	150	5
left_hip_yaw_joint	0	150	5
left_knee_joint	0.42	200	5
left_ankle_pitch_joint	-0.23	20	2
left_ankle_roll_joint	0	20	2
right_hip_pitch_joint	-0.2	200	5
right_hip_roll_joint	0	150	5
right_hip_yaw_joint	0	150	5
right_knee_joint	0.42	200	5
right_ankle_pitch_joint	-0.23	20	2
right_ankle_roll_joint	0	20	2
waist_yaw_joint	0	200	5
left_shoulder_pitch_joint	0.35	40	10
left_shoulder_roll_joint	0.16	40	10
left_shoulder_yaw_joint	0	40	10
left_elbow_joint	0.87	40	10
left_wrist_roll_joint	0	40	10
left_wrist_pitch_joint	0	40	10
left_wrist_yaw_joint	0	40	10
left_hand_index_0_joint	0	5	1.25
left_hand_index_1_joint	0	5	1.25
left_hand_middle_0_joint	0	5	1.25
left_hand_middle_1_joint	0	5	1.25
left_hand_thumb_0_joint	0	5	1.25
left_hand_thumb_1_joint	0	5	1.25
left_hand_thumb_2_joint	0	5	1.25
right_shoulder_pitch_joint	0.35	40	10
right_shoulder_roll_joint	-0.16	40	10
right_shoulder_yaw_joint	0	40	10
right_elbow_joint	0.87	40	10
right_wrist_roll_joint	0	40	10
right_wrist_pitch_joint	0	40	10
right_wrist_yaw_joint	0	40	10
right_hand_index_0_joint	0	5	1.25
right_hand_index_1_joint	0	5	1.25
right_hand_middle_0_joint	0	5	1.25
right_hand_middle_1_joint	0	5	1.25
right_hand_thumb_0_joint	0	5	1.25
right_hand_thumb_1_joint	0	5	1.25
right_hand_thumb_2_joint	0	5	1.25

TABLE VIII: Body joints grouped by body part.

Legs	
left_hip	right_hip
left_knee	right_knee
left_ankle	right_ankle
left_foot	right_foot
Spine & Torso	
pelvis	spine_1
spine_2	spine_3
neck	head
Arms	
left_collar	right_collar
left_shoulder	right_shoulder
left_elbow	right_elbow
left_wrist	right_wrist

For a given spatial control signal point at a given time-step and joint, it is considered functional only if the 3D spatial point is not (0,0,0). Hence, we initiate all spatial control points with (0,0,0) and then fill out values of joints at time-steps that we want to control.

TABLE VII: Open and Closed hand states for all tasks. Each config state for a hand consists of a tuple of 7 joint angles in the same order as in Table VIII. AOL: (0,0,0,0,0,0,0), ACL: $(-\pi/2, -\pi/2, -\pi/2, -\pi/2, 0, \pi/3, \pi/2)$, AOR: (0,0,0,0,0,0,0), ACR: $(\pi/2, \pi/2, \pi/2, \pi/2, 0, \pi/3, \pi/2)$, BPR: (0,0, $\pi/2$, $\pi/2$,0, $\pi/3$, $\pi/2$), DOR: (0, $\pi/2$,0, $\pi/2$,0,0,0)

Task	Left Hand Config (Open — Close)	Right Hand Config (Open — Close)	Group set to default q
Pick	ACL — ACL	AOR — ACR	$G^{\text{left arm}}$
Precise Punch	ACL — ACL	ACR — ACR	-
Precise Kick	ACL — ACL	ACR — ACR	-
Press Button	ACL — ACL	PBR — PBR	-
Jump	ACL — ACL	ACR — ACR	-
Sit	ACL — ACL	ACR — ACR	-
Bimanual Pick	ACL — ACL	ACR — ACR	-
Pick from Ground (side grasp)	AOL — ACL	ACR — ACR	$G^{\text{right arm}}$
Pick from Ground (top grasp)	ACL — ACL	ACR — ACR	$G^{\text{left arm}}$
Pick and Place	ACL — ACL	AOR — ACR	-
Open Drawer	ACL — ACL	DOR — ACR	-
Open Door	ACL — ACL	DOR — ACR	-

A. Task-specific prompts

All task-specific prompt texts, λ^{text} are listed in Table IX. The spatial control signals for each task are given as follows:-

1) *Pick*: For each reference trajectory:- We sample a target point, $p^{\text{target}} = \{p^x \in \mathcal{U}(1.0, 1.2), p^y \in \mathcal{U}(-0.4, 0.4), p^z = 1.1\}$. Target time-step is chosen to be, $t'_g = 50 + \lfloor 50(p^x - 1.0) \rfloor$. Then, we set spatial control signal for wrist as follows:-

$$\begin{aligned} \lambda_i^{\text{right.wrist}} &= p^{\text{target}} \forall i \in \{t'_g, \dots, t'_g + 20\} \\ \lambda_i^{\text{right.wrist}} &= (p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z} + 0.2) \\ &\forall i \in \{t'_g + 40, \dots, L\} \end{aligned}$$

We also set the target spatial points for elbow to encourage generating trajectories where the object is grabbed from side as follows:-

$$\begin{aligned} \lambda_i^{\text{elbow}} &= (p^{\text{target},x}, p^{\text{target},y} - 0.26 \cos(\frac{\pi}{4}), p^{\text{target},z} - 0.26 \sin(\frac{\pi}{4})) \\ &\forall i \in \{t'_g, \dots, t'_g + 20\} \end{aligned}$$

2) *Precise Punch*: For each reference trajectory:- We sample a target point, $p^{\text{target}} = \{p^x \in \mathcal{U}(1.2, 1.5), p^y \in \mathcal{U}(-0.2, 0.0), p^z = \mathcal{U}(1.0, 1.5)\}$. Target time-step is chosen to be, $t'_g = 30$. Then, we set spatial control signal for wrist as follows:-

$$\lambda_i^{\text{right.wrist}} = p^{\text{target}} \forall i \in \{t'_g - 10, \dots, t'_g + 10\}$$

3) *Precise Kick*: For each reference trajectory:- We sample a target point, $p^{\text{target}} = \{p^x \in \mathcal{U}(1.0, 1.2), p^y = 0.0, p^z = \mathcal{U}(0.5, 1.0)\}$. Target time-step is chosen to be, $t'_g = 30$. Then, we set spatial control signal for right foot as

TABLE IX: Text prompts for each task in simulation

Task	Prompts
Pick	"a person walks to cup, grabs the cup from side and lifts up"
Precise Punch	"a person performs a single boxing punch with his right hand"
Precise Kick	"a person stands and kicks with his right leg"
Press Button	"a person walks towards elevator, presses elevator button"
Jump	"a person jumps forward"
Sit	"a person walks towards a chair, sits down"
Bimanual Pick	"a person raises the toolbox with both hands"
Pick from Ground (Side Grasp)	"a person raises the toolbox with the use of one hand"
Pick from Ground (Top Grasp)	"a person walks forward, bends down to pick something up off the ground"
Pick and Place	"a person picks the cup and puts it on another table"

TABLE X: Text prompts and slow-down factor for each task deployed on real robot

Task	Prompts	Slow down factor
Pick	"a person stands in place, grabs the cup from side and lifts up"	2.5
Precise Punch	"a person performs a single boxing punch with his right hand"	1.5
Press Button	"a person stands in place, presses elevator button"	1.5
Bimanual Pick	"a person raises the toolbox with both hands"	1
Squat	"a person squats in place and stands up"	1

follows:-

$$\lambda_i^{\text{right-foot}} = p^{\text{target}} \forall i \in \{t'_g - 13, \dots, t'_g + 10\}$$

4) *Press Button*: For each reference trajectory:- We sample a target point, $p^{\text{target}} = \{p^x \in \mathcal{U}(1.4, 1.8), p^y = \mathcal{U}(-0.4, 0.4), p^z = \mathcal{U}(1.1, 1.2)\}$. Target time-step is chosen to be, $t'_g = 70 + \lfloor 50(p^x - 1.4) \rfloor$. Then, we set spatial control signal for right foot as follows:-

$$\begin{aligned} \lambda_i^{\text{right-wrist}} &= p^{\text{target}} \forall i \in \{t'_g, \dots, t'_g + 20\} \\ \lambda_i^{\text{right-wrist}} &= p^{\text{target},x} - 0.2, p^{\text{target},y}, p^{\text{target},z} \text{ for } i = t'_g + 40 \end{aligned}$$

5) *Jump*: We sample a trajectory with target point, $p^{\text{target}} = \{p^x = 1.0, p^y = 0.0, p^z = 1.9\}$. Target time-step is chosen to be $t'_g = 50$. We set the spatial control signal for pelvis as: $\lambda_i^{\text{pelvis}} = p^{\text{target}}$ for $i = t'_g$

6) *Sit*: We sample a trajectory with target point, $p^{\text{target}} = \{p^x = 1.3, p^y = 0.8, p^z = 0.58\}$. Target time-step is chosen to be $t'_g = 100$. We set the spatial control signal for pelvis as: $\lambda_i^{\text{pelvis}} = p^{\text{target}}$ for $i = t'_g$

7) *Bimanual Pick*: We use target point, $p^{\text{target}} = \{p^x = 0.7, p^y = 0.0, p^z = 0.65\}$. Target time-step is chosen to be $t'_g = 98$. We set the spatial control signal for left wrist as follows:-

$$\begin{aligned} \lambda_i^{\text{left-wrist}} &= \{p^{\text{target},x}, p^{\text{target},y} + 0.35, p^{\text{target},z} + 0.25 \\ &\quad - \frac{0.25i}{t'_g}\} \forall i \in \{0, \dots, t'_g\} \\ \lambda_i^{\text{left-wrist}} &= \{p^{\text{target},x}, p^{\text{target},y} + 0.15, p^{\text{target},z} - 0.25 \\ &\quad + \frac{0.25i}{t'_g}\} \forall i \in \{t'_g, \dots, t'_g + 98\} \end{aligned}$$

Spatial control signal for right wrist is as follows:-

$$\begin{aligned} \lambda_i^{\text{right-wrist}} &= \{p^{\text{target},x}, p^{\text{target},y} - 0.35, p^{\text{target},z} + 0.25 \\ &\quad - \frac{0.25i}{t'_g}\} \forall i \in \{0, \dots, t'_g\} \\ \lambda_i^{\text{right-wrist}} &= \{p^{\text{target},x}, p^{\text{target},y} - 0.15, p^{\text{target},z} - 0.25 \\ &\quad + \frac{0.25i}{t'_g}\} \forall i \in \{t'_g, \dots, t'_g + 98\} \end{aligned}$$

8) *Pick from Ground (Side Grasp)*: We use target point, $p^{\text{target}} = \{p^x = 0.5, p^y = 0.5, p^z = 0.1\}$. Target time-step is $t'_g = 98$. We set the spatial control signal for the left wrist as follows:-

$$\begin{aligned} \lambda_i^{\text{left-wrist}} &= \{p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z} + 0.5 - \frac{0.5i}{t'_g}\} \\ &\quad \forall i \in \{0, \dots, t'_g\} \\ \lambda_i^{\text{left-wrist}} &= \{p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z} - 0.5 + \frac{0.5i}{t'_g}\} \\ &\quad \forall i \in \{t'_g, \dots, t'_g + 98\} \end{aligned}$$

9) *Pick from Ground (Top Grasp)*: We use target point, $p^{\text{target}} = \{p^x = 1.0, p^y = 0.0, p^z = 0.2\}$. Target time-step is $t'_g = 50$. We set the spatial control signal for the right wrist as follows:-

$$\lambda_i^{\text{right-wrist}} = \{p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z}\} \text{ for } i = t'_g$$

10) *Pick and Place*: We use 2 target points, $p^{\text{target},1} = \{p^x = 1.2, p^y = -0.15, p^z = 0.6\}$ and $p^{\text{target},2} = \{p^x = 1.2, p^y = -0.6, p^z = 0.6\}$. Target time-steps are $t'_{g1} = 80$

and $t'_{g2} = 160$. We set the spatial control signal for the right wrist as follows:-

$$\begin{aligned}\lambda_i^{\text{right_wrist}} &= p^{\text{target},1} \text{ for } i = t'_{g1} \\ \lambda_i^{\text{right_wrist}} &= p^{\text{target},2} \text{ for } i = t'_{g2}\end{aligned}$$

11) Open Drawer: For drawer opening, the number of trajectories in HumanML dataset are very limited (only 18 with both “drawer” and “open” keywords in text description), due to which the generated motion was very off when prompted with drawer opening and only giving sparse spatial control signal for wrist at the drawer. To address this, we prompt DreamControl to generate 2 trajectories and no spatial control signal, one with prompt “stand still” and one with prompt “squat and stay in squat position”. We use the generated motion as initialization and solve for the following optimization problem with gradient descent from the current state. The target points are sampled as $p^{\text{target}} = \{p^x \in \mathcal{U}(0.3, 0.35), p^y \in \mathcal{U}(-0.2, -0.1), p^z \in \mathcal{U}(0.4, 0.8)\}$ and $t'_g = 40$ if $p^z \geq 0.7$ and $t'_g = 50$ if $p^z < 0.7$.

We define target trajectory, τ^{wrist} for wrist as follows if $p^z > 0.7$:-

$$\begin{aligned}\tau_i^{\text{wrist}} &= \{(p^{\text{target},x}) * \frac{i}{40}, -0.25 + (p^{\text{target},y} + 0.25) * \frac{i}{40}, \\ &\quad 0.7 + (p^{\text{target},z} - 0.7) * \text{quad}(\frac{i}{40})\} \\ \text{where } \text{quad}(x) &= 1 - (x - 1)^2, \forall i \in \{0, 1 \dots 40\} \\ \tau_i^{\text{wrist}} &= \{p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z}\} \forall i \in \{41, 42 \dots 70\} \\ \tau_i^{\text{wrist}} &= \{p^{\text{target},x} - \frac{0.2(i - 70)}{40}, p^{\text{target},y}, p^{\text{target},z}\} \\ &\quad \forall i \in \{71, 72 \dots 110\} \\ \tau_i^{\text{wrist}} &= \{p^{\text{target},x} - 0.2, p^{\text{target},y}, p^{\text{target},z}\} \forall i \in \{111, 112 \dots 196\}\end{aligned}$$

and for $p^z < 0.7$ where w_i is the wrist position in the original re-targeted trajectory.

$$\begin{aligned}\tau_i^{\text{wrist}} &= w_i \forall i \in \{0, 1, \dots, 50\} \\ \tau_{i+50}^{\text{wrist}} &= \{w_{50}^x + (p^{\text{target},x} - w_{50}^x) * \frac{i}{40}, w_{50}^y + (p^{\text{target},y} - w_{50}^y) * \frac{i}{40}, w_{50}^z + (p^{\text{target},z} - w_{50}^z) * \text{quad}(\frac{i}{40})\} \\ \text{where } \text{quad}(x) &= 1 - (x - 1)^2, \forall i \in \{0, 1, \dots, 40\} \\ \tau_{i+50}^{\text{wrist}} &= \{p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z}\} \forall i \in \{41, 42 \dots 70\} \\ \tau_{i+50}^{\text{wrist}} &= \{p^{\text{target},x} - \frac{0.2(i - 70)}{40}, p^{\text{target},y}, p^{\text{target},z}\} \\ &\quad \forall i \in \{71, 72, \dots, 110\} \\ \tau_{i+50}^{\text{wrist}} &= \{p^{\text{target},x} - 0.2, p^{\text{target},y}, p^{\text{target},z}\} \\ &\quad \forall i \in \{111, 112, \dots, 146\}\end{aligned}$$

With this target trajectory, we run gradient descent to optimize only the joint angles q_i^{ref} with the following loss in order to align the wrist to the target trajectory:-

$$\begin{aligned}L &= \sum_{i=0}^{L-1} (p_i^{\text{wrist}} - w_i)^2 \\ \text{where } p_i^{\text{wrist}} &= f k^{\text{wrist}}(p_i^{\text{ref,root}}, \theta_i^{\text{ref,root}}, q_i^{\text{ref,*}}) \\ q_i^{\text{ref,j+1}} &= q_i^{\text{ref,j}} - \lambda^{\text{lr}} \frac{\partial}{\partial q_i^{\text{ref,*}}} L\end{aligned}$$

12) Open Door: We generate trajectories for door opening in a similar way to drawer opening with the only changes in the target trajectory as follows for $p^{\text{target},z} \geq 0.7$:-

$$\begin{aligned}\tau_i^{\text{wrist}} &= \{(p^{\text{target},x}) * \frac{i}{40}, -0.25 + (p^{\text{target},y} + 0.25) * \frac{i}{40}, \\ &\quad 0.7 + (p^{\text{target},z} - 0.7) * \text{quad}(\frac{i}{40})\} \\ \text{where } \text{quad}(x) &= 1 - (x - 1)^2, \forall i \in \{0, 1 \dots 40\} \\ \tau_i^{\text{wrist}} &= \{p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z}\} \forall i \in \{41, 42 \dots 70\} \\ \tau_i^{\text{wrist}} &= \{c^x - R \sin(a_i) - h \cos(a_i), c^y + R \cos(a_i) \\ &\quad - h \sin(a_i), p^{\text{target},z}\} \\ \forall i \in \{71, 72 \dots 110\} &\text{ where } c^x = p^{\text{target},x} + h, \\ c^y &= p^{\text{target},y} - R, a_i = \frac{i - 70}{40} \\ \tau_i^{\text{wrist}} &= \{p^{\text{target},x} - 0.2, p^{\text{target},y}, p^{\text{target},z}\} \forall i \in \{111, 112 \dots 196\}\end{aligned}$$

and for $p^{\text{target},z} < 0.7$:-

$$\begin{aligned}\tau_i^{\text{wrist}} &= w_i \forall i \in \{0, 1, \dots, 50\} \\ \tau_{i+50}^{\text{wrist}} &= \{w_{50}^x + (p^{\text{target},x} - w_{50}^x) * \frac{i}{40}, w_{50}^y + (p^{\text{target},y} - w_{50}^y) * \frac{i}{40}, w_{50}^z + (p^{\text{target},z} - w_{50}^z) * \text{quad}(\frac{i}{40})\} \\ \text{where } \text{quad}(x) &= 1 - (x - 1)^2, \forall i \in \{0, 1, \dots, 40\} \\ \tau_{i+50}^{\text{wrist}} &= \{p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z}\} \forall i \in \{41, 42 \dots 70\} \\ \tau_{i+50}^{\text{wrist}} &= \{c^x - R \sin(a_i) - h \cos(a_i), c^y + R \cos(a_i) \\ &\quad - h \sin(a_i), p^{\text{target},z}\} \\ \forall i \in \{71, 72 \dots 110\} &\text{ where } c^x = p^{\text{target},x} + h, \\ c^y &= p^{\text{target},y} - R, a_i = \frac{i - 70}{40} \\ \tau_{i+50}^{\text{wrist}} &= \{p^{\text{target},x} - 0.2, p^{\text{target},y}, p^{\text{target},z}\} \\ &\quad \forall i \in \{111, 112, \dots, 146\}\end{aligned}$$

B. Trajectory filtering

As DreamControl does not consider scene elements to avoid collision with like the platform on which the target object is kept, some of the generated trajectories turn out to be infeasible to be used for reference trajectory in stage 2. Also, some of the generated trajectories empirically turned out to be very dynamically infeasible in practice to be tracked by RL in stage 2. Hence, we filter out trajectories based on some heuristics like reject a trajectory if it collides with the scene environment, reject if it bends it's waist too much etc.

TABLE XI: # of filtered trajectories and constants used for filtering

Task	#Trajs before	#Trajs after	β_{torso}	β_{pelvis}
Pick	100	67	$\pi/4$	0.6
Precise Punch	100	100	$\pi/4$	0.6
Precise Kick	100	66	$\pi/2$	0.5
Press Button	100	96	$\pi/3$	0.5

in the trajectory. Specifically, these are the conditions used to filter out trajectories:-

- Reject if torso angle with x axis is larger than β_{torso} i.e. reject if $\arccos(\text{axis}_x^{\text{torso},x}) > \beta_{\text{torso}}$ where $\text{axis}_z^{\text{torso}}$ is the x axis of the torso. This is to reject reference trajectories where the robot bends too much or turns around which is undesired for tasks in question
- Reject if pelvis height is below a certain threshold β_{pelvis} . This is to reject reference trajectories where robot squats too much for tasks it is not needed to.
- Reject if any part of the body collides with the scene

The specific thresholds, β_{torso} and β_{pelvis} for tasks that we do filtering and number of filtered trajectories from total sampled are given in Table XI. For all other tasks except Open Drawer and Open Door, only 1 trajectory is manually selected for training which can be easily scaled up with careful engineering for their automated filtering. For Open Drawer and door tasks, as trajectories are obtained by optimization of right arm, it does not need filtering. It must be noted that these heuristic-based filtering is only required because diffusion-generated trajectories are unfit for tracking. This maybe mainly because diffusion model sees out-of-distribution samples during annealing. With more data, the need for filtering out samples maybe eliminated.

C. Trajectory refinement

The trajectories obtained after re-targeting and filtering do not start from a same pose. This is a characteristic of DreamControl that it only restricts the $p^{\text{ref root}, x}$, $p^{\text{ref root}, y}$ and θ^{yaw} to start from origin at $t = 0$ but the generated motion could start from any pose at $t = 0$. However, for RL training in stage 2, we need reference trajectories to start from a fixed pose and point. Hence, in each generated reference trajectory, we prepend a trajectory that starts from a fixed default joint pose at fixed pose of the root and interpolates to the start joint pose and root pose of the generated trajectory. Specifically, we append $N^{\text{init}} = 20$ frames at the beginning of each motion of which for the first 10 frames the trajectory remains static and next 10 frames to interpolate to α_0 , to obtain refined reference trajectory, α^{refined} consisting of 216 frames.

Further, we refine motions by disabling movement of the non-functional arm based on the task to avoid it's unnecessary movement in the reference trajectory. Specifically, we set the joint angles of the left arm and right arm group joints denoted as $G^{\text{left arm}}$ and $G^{\text{right arm}}$ as in Table VII to their default values along all time steps. The default joint angles. The default values are given in Table VI. The specific group that was refined to set to default joint angles for each task

is given in Table VII. The tasks not included in table do not have this refinement applied.

1) *Special case for Pick task:* Last, specifically for *Pick* task, we observed that as our robot is shorter (1.32m) than the SMPL model used to train generative model on (1.74m), most motions in the dataset are of the human picking object from platform which is located near or below it's waist while for the shorter robot, the platform is more close to it's shoulder above it's waist. Hence, most generated motion when the target point is higher for go through the platform. Hence, specifically for the *Pick* task, we add an additional refinement layer before passing them for filtering to minimally modify the reference trajectory to avoid collision of right hand with the platform. Specifically, we solve the following optimization problem through gradient descent with joint angle variables initialized from the reference trajectory:-

$$q^{\text{ref}*} = \underset{q}{\text{argmin}} \sum_{t=\{\Delta t, \dots, (L-1)\Delta t\}} (||p_t^{\text{right hand},*} - p_{t-1}^{\text{right hand},*}||_2 - ||p_t^{\text{right hand}} - p_{t-1}^{\text{right hand}}||_2) \\ \text{s.t. } d(p_t^{\text{right hand}}) = 0$$

where $d(\cdot)$ is a function that maps 3D points to their closest distance from the free space. The rationale for choosing relative distance difference from reference as the objective is to preserve the smoothness of the motion in the generated trajectory but just modify the trajectory minimally to avoid collision of right hand with the platform. That is if the reference trajectory was to slow down while approaching the object as humans do, that slowing down will still be preserved in the trajectory after modifying it to avoid collision. If after making this refinement, if some other body part like feet in the reference trajectory makes collision with the platform, that trajectory is rejected.

D. Computation requirements

All training was conducted on NVIDIA A6000 GPUs with 48GB VRAM. Training took about 1 hr of clock time for each task (for both privileged simulation and Sim2Real training). It took about 15 mins to get reference trajectories from OmniControl, clean and refine them specific to the task.

E. Sim2Real

For sim2real trajectory generation and refinement, we follow a similar design to their corresponding sim tasks with some changes in prompt and trajectory refinements that we will discuss in this section. There are 5 tasks that we put on hardware robot. Specific prompts used for them are listed in Table X. The generated trajectories are also slowed down by a certain factor as also reported in Table X. This is to ensure safety of the motion and executed and minimize the sim-real gap by making the motion slower. It was empirically observed that with the same speed, due to some sim-to-real gap, the executed trajectories exhibited less success rates for precise tasks like pick where the object got pushed a little bit and thus failed to be grasped as the position is not updated

in the current design. This slowing down was not required for Squat and Bimanual Pick tasks. In addition to the standard refinements as made in the sim versions of these tasks (except Squat), the specific spatial control points and refinements for each task are as follows:-

1) *Pick*: For each reference trajectory:- We sample a target point, $p^{\text{target}} = \{p^x \in \mathcal{U}(0.25, 0.35), p^y \in \mathcal{U}(-0.3, 0.0), p^z = 1.1\}$. Target time-step is chosen to be, $t'_g = 30$. Then, we set spatial control signal for wrist as follows:-

$$\begin{aligned}\lambda_i^{\text{right_wrist}} &= p^{\text{target}} \forall i \in \{t'_g, \dots, t'_g + 20\} \\ \lambda_i^{\text{right_wrist}} &= (p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z} + 0.2) \\ &\quad \forall i \in \{t'_g + 20, \dots, t'_g + 40\}\end{aligned}$$

We also set the target spatial points for elbow to encourage generating trajectories where the object is grabbed from side as follows:-

$$\begin{aligned}\lambda_i^{\text{elbow}} &= (p^{\text{target},x}, p^{\text{target},y} - 0.26 \cos(\frac{\pi}{4}), p^{\text{target},z} - 0.26 \sin(\frac{\pi}{4})) \\ &\quad \forall i \in \{t'_g, \dots, t'_g + 20\}\end{aligned}$$

Of the generated trajectories, we refine them to avoid collision with the platform, plus we also add an additional cost to bring the trajectories close to their corresponding goal points for which the trajectories were generated.

$$\begin{aligned}q^{\text{ref},*} &= \underset{t=\{\Delta t, \dots, (L-1)\Delta t\}}{\text{argmin}} (| \|p_t^{\text{right hand},*} - p_{t-1}^{\text{right hand},*} \|_2 \\ &\quad - \|p_t^{\text{right hand}} - p_{t-1}^{\text{right hand}} \|_2 |) \\ &\quad + M_t \|p_t^{\text{right wrist}} - \lambda_t^{\text{right wrist}} \| \\ \text{s.t. } d(p_t^{\text{right hand}}) &= 0\end{aligned}$$

Where $M_t = 1$ for $t \in \{0, t'_g, t'_g + 1, \dots, t'_g + 40\}$ and 0 otherwise. This is to mask for timesteps where the goal is specified. Idea is to minimally modify the trajectories such that they follow the corresponding goal. This refinement was required to ensure smooth policies. We also set the lower body (groups $G^{\text{left leg}}$ and $G^{\text{right leg}}$) to fixed default joint angles and adjust the root height such that the feet touches the ground. This is to enforce that the motion is stand still.

2) *Precise Punch*: For each reference trajectory:- We sample a target point, $p^{\text{target}} = \{p^x \in \mathcal{U}(0.6, 0.65), p^y \in \mathcal{U}(-0.2, 0.1), p^z = \mathcal{U}(1.15, 1.4)\}$. Target time-step is chosen to be, $t'_g = 30$. Then, we set spatial control signal for wrist as follows:-

$$\begin{aligned}\lambda_i^{\text{right_wrist}} &= p^{\text{target}} \forall i \in \{t'_g, \dots, t'_g + 20\} \\ \lambda_i^{\text{right_wrist}} &= (p^{\text{target},x}, p^{\text{target},y}, p^{\text{target},z} + 0.2) \\ &\quad \forall i \in \{t'_g + 20, \dots, t'_g + 40\}\end{aligned}$$

Of the generated trajectories, we refine them to bring the trajectories close to their corresponding goal points for which the trajectories were generated.

$$\begin{aligned}q^{\text{ref},*} &= \underset{t=\{\Delta t, \dots, (L-1)\Delta t\}}{\text{argmin}} (| \|p_t^{\text{right hand},*} - p_{t-1}^{\text{right hand},*} \|_2 \\ &\quad - \|p_t^{\text{right hand}} - p_{t-1}^{\text{right hand}} \|_2 |) \\ &\quad + M_t \|p_t^{\text{right wrist}} - \lambda_t^{\text{right wrist}} \| \\ \text{s.t. } d(p_t^{\text{right hand}}) &= 0\end{aligned}$$

Where $M_t = 1$ for $t \in \{0, t'_g\}$ and 0 otherwise. This is to mask for timesteps where the goal is specified. Idea is to minimally modify the trajectories such that they follow the corresponding goal. This refinement was required to ensure smooth policies. We also set the lower body (groups $G^{\text{left leg}}$ and $G^{\text{right leg}}$) to fixed default joint angles and adjust the root height such that the feet touches the ground. This is to enforce that the motion is stand still.

3) *Bimanual Pick*: We follow the exact same spatial control points as the sim to generate the trajectory. We additionally refine to dismiss any feet slipping by using IK to adjust $G^{\text{left leg}}$ and $G^{\text{right foot}}$ to bring left and right foot to their positions at $p_0^{\text{left foot}}$ and $p_0^{\text{right foot}}$. Other than this, we set the *roll* and *yaw* of root to 0 for symmetry.

4) *Open Drawer*: We follow the exact same procedure as the open drawer trajectories for sim.

5) *Squat*: For each trajectory, we sample a trajectory with target point, $p^{\text{target}} = \{p^x = -0.15, p^y = 0.0, p^z = \mathcal{U}(0.4, 0.6)\}$. Target time-step is chosen to be $t'_g = 100$. We set the spatial control signal for pelvis as: $\lambda_i^{\text{pelvis}} = p^{\text{target}}$ for $i = t'_g$. We additionally refine to dismiss any feet slipping by using IK to adjust $G^{\text{left leg}}$ and $G^{\text{right foot}}$ to bring left and right foot to their positions at $p_0^{\text{left foot}}$ and $p_0^{\text{right foot}}$. Other than this, we set the *roll* and *yaw* of root to 0 for symmetry.

X. RL TRAINING

A. Model architecture

We use a simple fully-connected MLP to represent both our policy (or actor) and critic for each task. The network architecture for the actor and critic has the following hidden layers: (512, 256, 256). Also, we use the same observations for policy and critic as opposed to asymmetric actor-critic setup in [73].

B. Environment parameters

We randomize the object/target location (by sampling different trajectories in some tasks), object mass, friction of surface/object. All of these variations for each task are listed in Table XII.

C. Task-specific sparse rewards

We add task-specific sparse rewards that are chosen to be indicative of whether the task is successful as well. These rewards are listed in Table XIII. For each task, we have a body part link, b that maybe used in the definition of the sparse rewards and the time t_g when the interaction is supposed to happen.

TABLE XII: Environment randomization parameters for each task

Task	Friction	Mass of object
Pick	$\mathcal{U}(0.7, 1)$	$\mathcal{U}(0.1, 1)$
Precise Punch	$\mathcal{U}(0.7, 1)$	-
Precise Kick	$\mathcal{U}(0.7, 1)$	-
Press Button	$\mathcal{U}(0.7, 1)$	-
Jump	$\mathcal{U}(0.7, 1)$	-
Sit	$\mathcal{U}(0.7, 1)$	-
Bimanual Pick	$\mathcal{U}(0.7, 1)$	$\mathcal{U}(0.1, 5)$
Pick from Ground (Side Grasp)	$\mathcal{U}(0.7, 1)$	$\mathcal{U}(0.1, 1)$
Pick from Ground (Top Grasp)	$\mathcal{U}(0.7, 1)$	$\mathcal{U}(0.1, 0.5)$
Pick and Place	$\mathcal{U}(0.7, 1)$	$\mathcal{U}(0.1, 0.5)$
Open Drawer	$\mathcal{U}(0.7, 1)$	-
Open Door	$\mathcal{U}(0.7, 1)$	-

TABLE XIII: Task-specific sparse rewards for each task

Task	Body part link, b	Sparse reward, r_t^{sparse}	Description
Pick	right_wrist_yaw.link	$1_{h_t^{\text{object}} > h_t^{\text{thres}}} 1_{t \geq t_b}$	$h_t^{\text{thres}} = 0.95$, h_t^{object} is height of object
Precise Punch	right_wrist_yaw.link	$1_{\ p_t^b - p^{\text{goal}}\ _2 < d^{\text{thres}}} 1_{t \geq t_b - 0.1} 1_{t \leq t_b + 0.1}$	$d^{\text{thres}} = 0.05$
Precise Kick	right_ankle_roll.link	$1_{\ p_t^b - p^{\text{goal}}\ _2 < d^{\text{thres}}} 1_{t \geq t_b - 0.1} 1_{t \leq t_b + 0.1}$	$d^{\text{thres}} = 0.1$
Press Button	right_wrist_yaw.link	$1_{\ p_t^b - p^{\text{goal}}\ _2 < d^{\text{thres}}} 1_{t \geq t_b - 0.1} 1_{t \leq t_b + 0.1}$	$d^{\text{thres}} = 0.05$
Jump	pelvis	$1_{\ p_t^b - p^{\text{goal}}\ _2 < d^{\text{thres}}} 1_{t \geq t_b - 0.1} 1_{t \leq t_b + 0.1}$	$d^{\text{thres}} = 0.1$
Sit	pelvis	$1_{\ p_t^b - p^{\text{goal}}\ _2 < d^{\text{thres}}} 1_{t \geq t_b}$	$d^{\text{thres}} = 0.05$
Bimanual Pick	right_wrist_yaw.link, left_wrist_yaw.link	$1_{h_t^{\text{object}} > h_t^{\text{thres}}} 1_{t \geq t_b}$	$h_t^{\text{thres}} = 0.7$, h_t^{object} is height of object
Pick from Ground (Side Grasp)	left_wrist_yaw.link	$1_{h_t^{\text{object}} > h_t^{\text{thres}}} 1_{t \geq t_b}$	$h_t^{\text{thres}} = 0.2$, h_t^{object} is height of object
Pick from Ground (Top Grasp)	right_wrist_yaw.link	$1_{h_t^{\text{object}} > h_t^{\text{thres}}} 1_{t \geq t_b}$	$h_t^{\text{thres}} = 0.3$, h_t^{object} is height of object
Pick and Place	right_wrist_yaw.link	$1_{\ p_t^{\text{object}} - p^{\text{goal}}\ _2 < d^{\text{thres}}} 1_{t \geq t_b}$	$d^{\text{thres}} = 0.1$, p^{goal} is position of the goal
Open Drawer	right_wrist_yaw.link	$1_{a_t^{\text{drawer}} > a_t^{\text{thres}}} 1_{t \geq t_b}$	a_t^{drawer} is drawer open amount, $a_t^{\text{thres}} = 0.05$
Open Door	right_wrist_yaw.link	$1_{a_t^{\text{door}} > a_t^{\text{thres}}} 1_{t \geq t_b}$	a_t^{door} is door open amount, $a_t^{\text{thres}} = 0.05$

D. Task-specific dense rewards for TaskOnly+

We also add a task-specific dense reward to encourage pre-grasp/pre-approach pose for the object/goal respectively. This task-specific reward is adapted from [8] and is defined as follows:-

$$r_t^{\text{dense}} = \|p_{t-\Delta t}^b - p_{t_g}^{\text{ref},b}\|_2 - \|p_t^b - p_{t_g}^{\text{ref},b}\|_2.$$

The specific body parts used in each task are listed in Table XIII.

E. Reward weights for each task

The rewards weights for each task are listed in Table XIV.

TABLE XIV: Reward weights for all tasks

Task	Tracking					Smoothness					$w_{\tau,\text{task,sparse}}$	$w_{\tau,\text{task,dense}}$
	w_{r_1}	w_{r_2}	w_{r_3}	w_{r_4}	w_{r_5}	w_{r_6}	w_{r_7}	w_{r_8}	w_{r_9}	$w_{r_{10}}$		
Pick	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	-0.5	-1	0.1	100
Precise Punch	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	-0.5	-1	1	100
Precise Kick	-0.2	-0.1	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	-0.15	-1 for left, -0.3 for right	1	100
Press Button	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	-0.5	-1	1	100
Jump	-0.2	-0.1	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	0	-1	1	100
Sit	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	-0.5	-1	1	100
Bimanual Pick	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	0	-1	0.1	100
Pick from Ground (Side Grasp)	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	-0.5	-1	0.1	100
Pick from Ground (Top Grasp)	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	-0.15	-1	0.1	100
Pick and Place	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	-0.5	-1	0.1	100
Open Drawer	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	0	-1	0.1	100
Open Door	-0.2	-0.05	-0.2	-0.2	0.3	$-1.5e^{-7}$	$-5e^{-3}$	-0.1	0	-1	0.1	100